

8

Forward Curve Models

8.1 Introduction

Historically the majority of work on modelling energy and commodity prices has been focussed on stochastic processes for the spot price and other key variables, such as the convenience yield and interest rates (examples include; Schwartz (1997), Gibson and Schwartz (1990), Hilliard and Reis (1998), Miltersen and Schwartz (1998)). However, this approach has some fundamental disadvantages – firstly the key state variables, such as the convenience yield, are unobservable and secondly the forward price curve is an endogenous function of the model parameters and therefore will not necessarily be consistent with the market observable forward prices. Figure 8.1, which reproduces figure 4.1, illustrates three market forward energy curves. Each curve is a plot of the first 24 monthly contracts with prices taken from NYMEX. The two crude oil curves (left-hand scale) show that the forward curve moved from backwardation into contango over the period 1 March 1999 to 31 March 2000. The third curve is the Henry Hub natural gas contract (right hand scale) observed on the 31 March 2000 and clearly shows the strong seasonality of natural gas prices associated with the Northern hemisphere's seasons.

For derivative pricing, therefore, many industry participants require the forward curve to be an *input* into the derivative pricing model, rather than an *output* from it, as is the case with the constant parameter versions of the spot price models in chapter 6. An approach based on modelling the entire forward price curve with multiple sources of uncertainty was introduced by Jamshidian (1991b) and used in Cortazar and Schwartz (1994) to analyse the pricing of Copper Interest-Indexed Notes¹. This approach uses all the information contained in the term structure of futures prices in addition to the historical volatilities of futures returns for different maturities. A similar approach has been put forward by Joy (1998a, 1999b) in which the author adds jumps modelled as a Poisson process. The model that he proposes is applied to the power markets and the jumps are assumed to have limited duration to reflect their transient nature in the markets. The framework of this chapter is based on these approaches and draws on material from the paper of Clewlow and Strickland (1999b)². In this chapter we show how the multi-factor model can be developed in stages and we then apply it to oil, gas and electricity markets. We show how the multi-factor model can be calibrated to observable market data, and derive formulae for the pricing of standard energy

¹ See also Amin, et al (1995).

² See also Clewlow and Strickland (1999c).

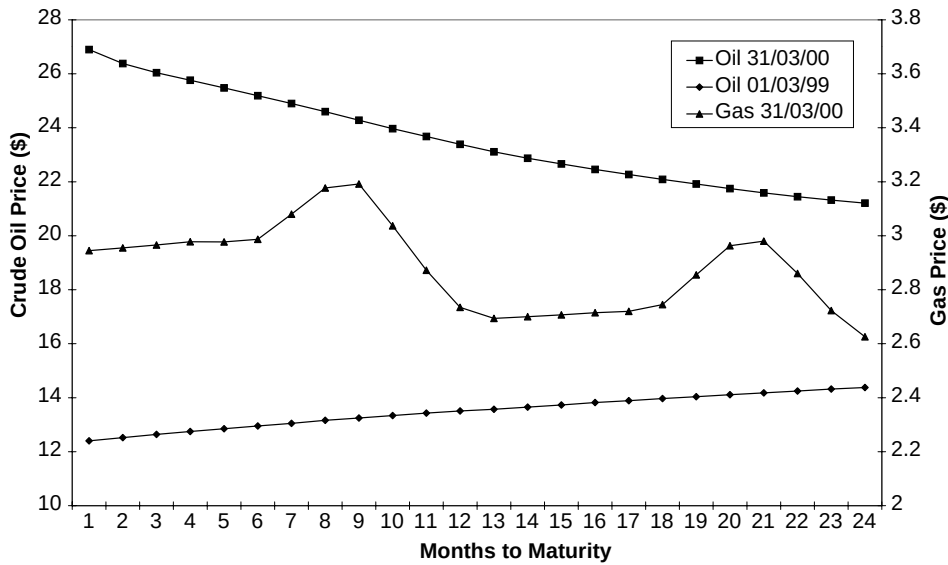


FIGURE 8.1 Market Energy Forward Curves

derivatives. These pricing formulae are then extended to enable the pricing of path dependent derivatives. We also discuss how the model can be adapted to handle seasonal volatility and volatility smiles.

8.2 A Simple Model for the Forward Curve

We will use the expression forward curve models to mean models that explicitly model all the forward energy prices simultaneously rather than just the spot price as with the models in chapters 6 and 7. To illustrate, we begin with a simple single factor model of the forward curve which we represent by the following stochastic differential equation;

$$\frac{dF(t, T)}{F(t, T)} = \sigma e^{-\alpha(T-t)} dz(t) \quad (8.1)$$

The inputs to the model are the observed forward curve $F(t, T)$ which denotes the forward price at time t for maturity date T , and $\sigma e^{-\alpha(T-t)}$ which is the single “factor” or volatility function associated with the source of risk $dz(t)$. Notice that equation (8.1) has no drift term – because futures and forward contracts have zero initial investment, their expected return in a risk-neutral world must be zero, implying that the process describing

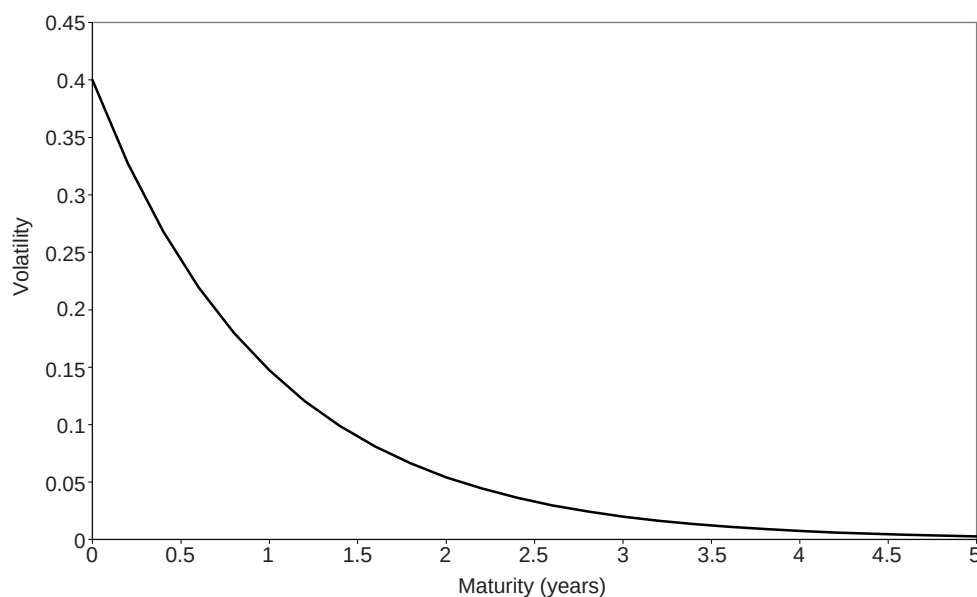


FIGURE 8.2 A Simple Negative Exponential Volatility Function for Forward Prices

their evolution has zero drift³. The volatility function of equation (8.1) has a very simple negative exponential form which we illustrate in figure 8.2.

For this volatility function short dated forward returns are more volatile than long dated forwards – information occurring in the market today has little effect on, say, the 5 year forward price but can have a significant effect on the one month forward price. The parameter values used for the figure are $\alpha = 1.0$ and $\sigma = 0.40$. Here σ represents the ‘overall’ volatility of the forward curve whilst α tells us how fast the forward volatility curve attenuates with increasing maturity. With an α of 100% we see that the one month forward has a volatility of about 37%, decreasing to roughly 2% for the 3 year forward.

Figure 8.3 shows a graphical representation of the evolution of the forward curve described by equation (8.1) with a volatility function of this type. The lower curve, denoted by $F(t, T)$, is the original observed market forward curve, whilst the upper curve, $F(t + dt, T)$, represents the curve after a small time step dt where there has been a positive shock to the system ($dz(t) > 0$). In this case the whole forward curve shifts up, with each point a multiple of $\sigma e^{-\alpha(T-t)}$ of the shock.

The volatility function is not restricted to have the parameterised form of equation

³ If we wish to work with the real world process we can add a drift to the models in this chapter without affecting the modelling results. However, for derivative pricing we will work in a risk neutral world.

(8.1). Instead we can allow the function to be very general;

$$\frac{dF(t, T)}{F(t, T)} = \sigma(t, T)dz(t) \quad (8.2)$$

where $\sigma(t, T)$ would be read as ‘the time t volatility of the T -maturity forward price return’. We can determine from market data what the form of $\sigma(t, T)$ should be. As we shall see later in the chapter we can also allow the forward curve to be affected by multiple sources of uncertainty.

8.3 The Dynamics of the Forward Curve

In order to determine empirically the form of the volatility function(s) we can look at the historical evolution of the market forward data. Figures 8.4 and 8.5 show the evolution of the NYMEX oil and natural gas futures curves respectively between January and June 1999 for the first 24 monthly contracts. For each figure the x -axis displays the maturity structure of the forward curve, the y -axis the date on which the curve is observed and the z -axis the market level of the futures price.

Both the oil and gas forward curves can be seen to move in a similar way to that illustrated by figure 8.3 with the shorter maturity forwards being more volatile than the longer maturity forwards. However, it is also clear that the real movement of the forward

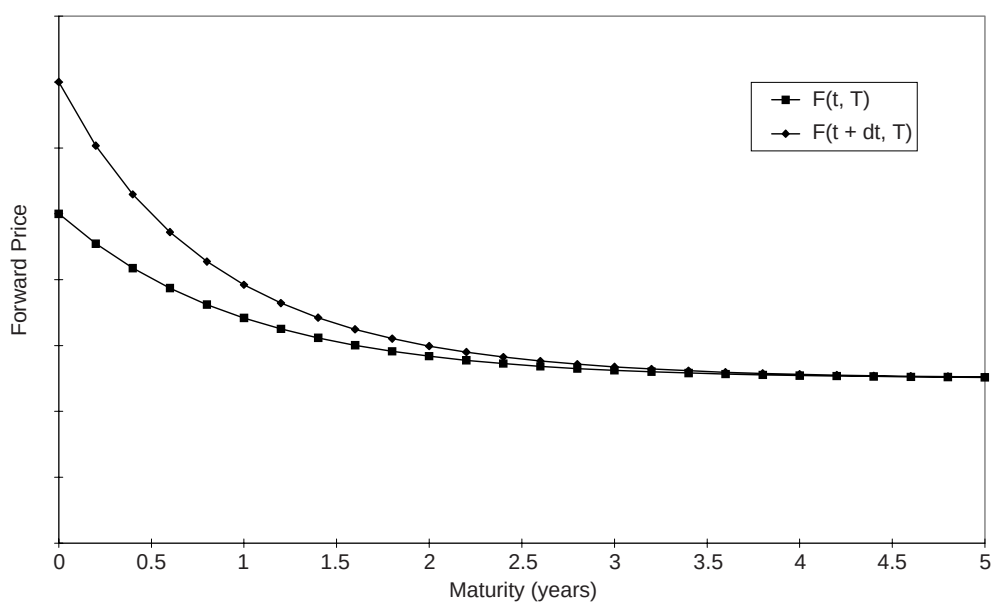


FIGURE 8.3 Example Evolution of the Forward Price Curve

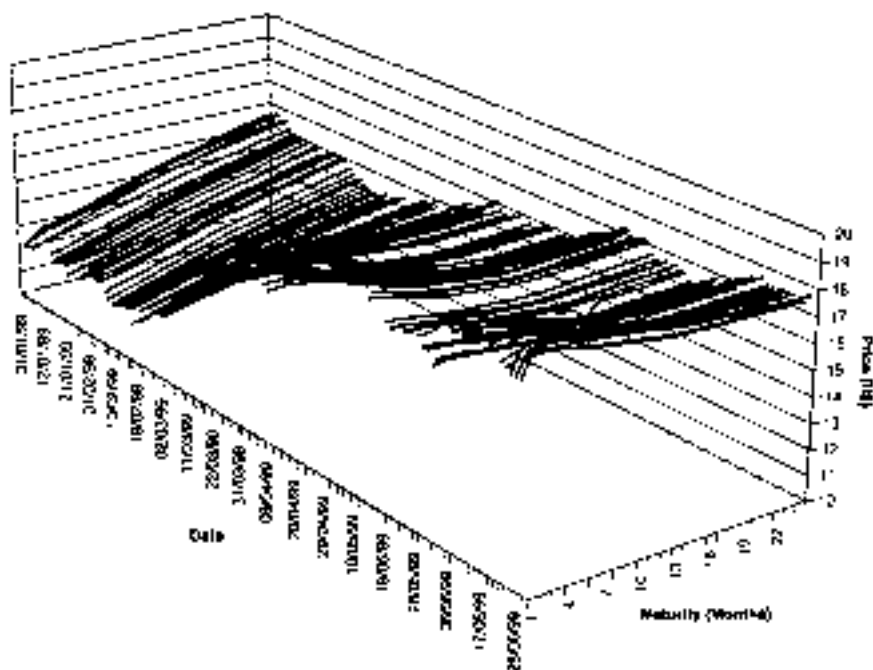


FIGURE 8.4 Evolution of NYMEX Oil Data (January 1999 to June 1999)

curve is probably more complex than is represented by equation (8.1). In particular, the seasonality in the gas forward curve can be clearly seen in figure 8.5. By looking at the annualised standard deviations of the forwards returns for each maturity, we can obtain an estimate of the overall volatility structure of the forward curve. Figure 8.6 shows the annualised standard deviation of returns for both oil and gas for the monthly contracts going out 2 years using daily data over the period from April 1998 to March 2000.

These curves can be used as the general volatility function $\sigma(t, T)$ in equation (8.2). Figure 8.6 shows empirically that short maturity forward prices are more volatile than longer maturity forward prices. Although volatility declines with maturity we see that it levels off at a non-zero level. In Figure 8.7 we fit the simple negative exponential function of equation (8.1) to the first six months of the empirically observed natural gas volatility curve by minimising the sum of the squared differences between the two curves. We see that, in order to capture the strong mean reversion implied in the declining volatility for the short dated contracts (the implied mean reversion rate is 1.9), the simple model forces the volatility for longer maturity forwards to go to zero too quickly. This leads the model to underestimate the volatility of the longer maturity gas forward prices and so would give a downward bias to option values that depend on this part of the curve.

An important observation from Figures 8.4 and 8.5 is that forward prices of different maturities are not perfectly correlated – the curves generally move up and down together

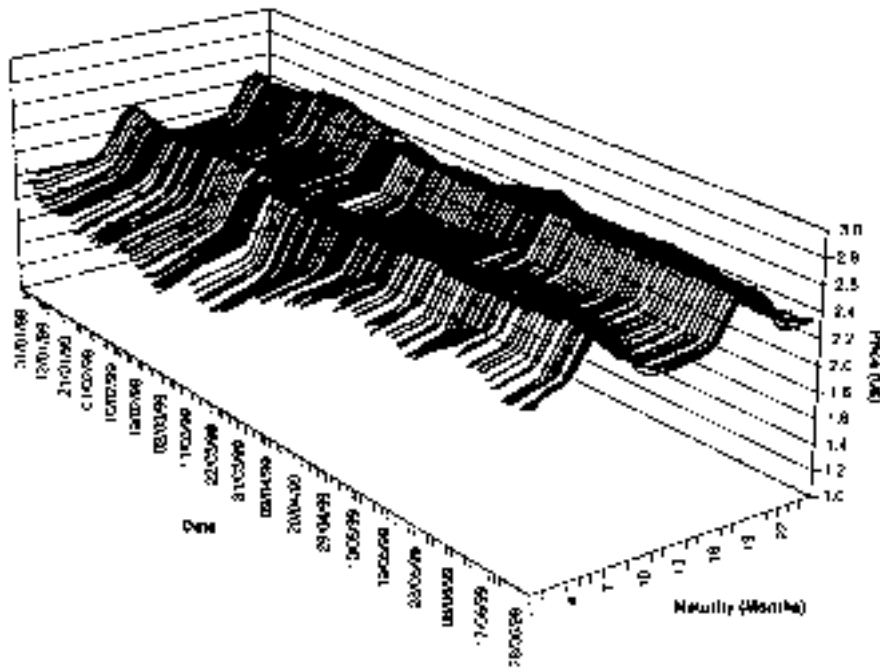


FIGURE 8.5 Evolution of NYMEX Natural Gas Data (January 1999 to June 1999)

but they also change shape in apparently quite complex ways. One method that can be used to determine the set of common factors that drive the dynamics of the forward curve is principal components analysis (PCA) or eigenvector decomposition of the covariance matrix. This procedure can be utilised to simultaneously identify the number of important factors and estimate the volatility functions. The technique involves calculating the covariances between every pair of forward price returns in an historical time series, similar to those shown in figures 8.4 and 8.5, to form a covariance matrix. The eigenvectors of the covariance matrix yield estimates of the factors driving the evolution of the forward curve. We will give a detailed example of how to perform this analysis in section 8.6.

As we shall see later these factors can be used as the volatility functions in a multi-factor model of the forward curve. Figure 8.8 shows the three most important volatility functions for the NYMEX crude oil futures curve using data from April 1998 to March 2000.

Figure 8.8 shows the typical pattern obtained from PCA – generally there are risk factors which act to “shift”, “tilt” and “bend” the forward curve. The most important factor “fn 01”, is positive for all maturities, implying that a positive shock to the system causes all prices to “shift” up (although by slightly different amounts). The second most important factor, “fn 02”, is a “tilt” factor which causes the short and long maturity

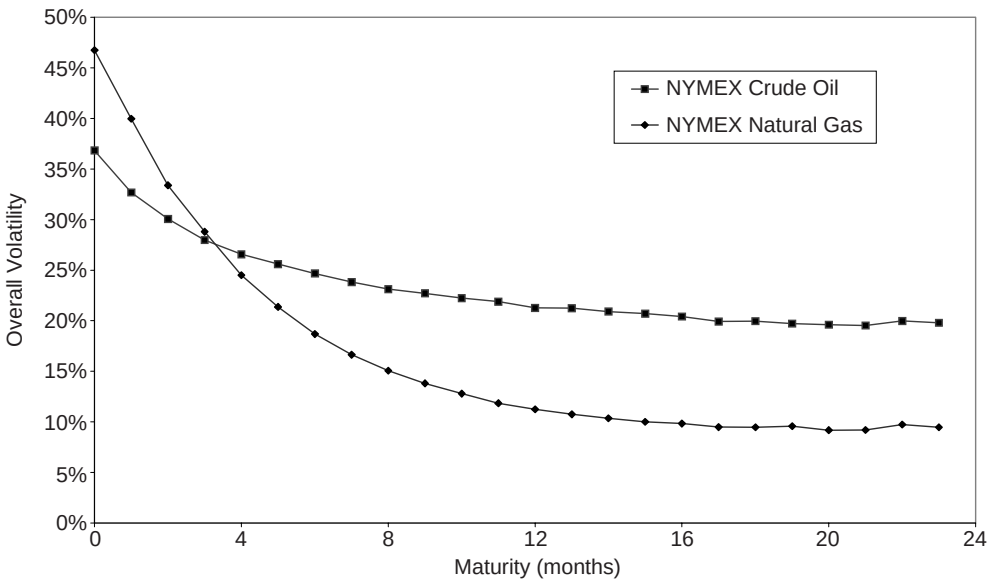


FIGURE 8.6 Overall Volatilities for NYMEX Crude Oil and Gas Futures Contracts

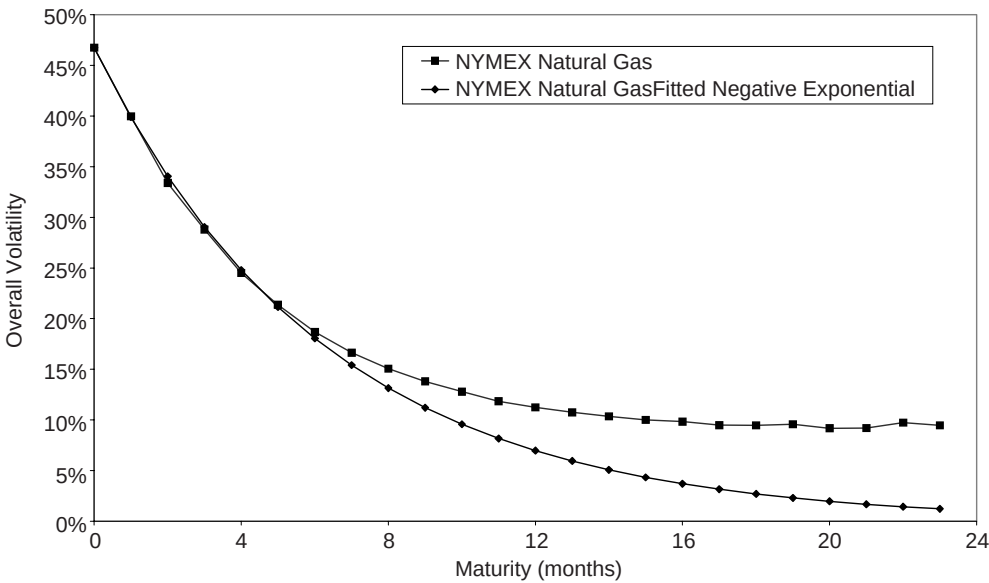


FIGURE 8.7 NYMEX Natural Gas Market and Fitted Model Volatilities

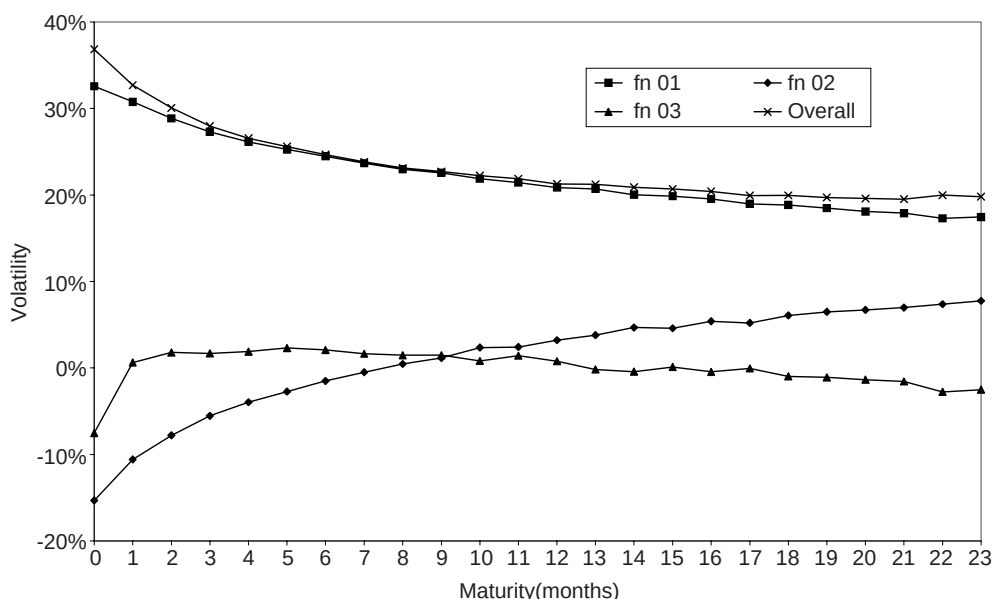


FIGURE 8.8 Three Main Volatility Functions for NYMEX Crude Oil Data From April 1998 to March 2000

ends of the curve to move in opposite directions. The third factor, “fn 03”, is a “bending” factor which causes the short and long ends to move in the opposite direction to the middle section. The “overall” volatility, which is the combined effect of all the three factors, is also shown in figure 8.8.

For many energies, as we have seen for natural gas and electricity, seasonality in the forward price volatilities is an important feature in the evolution of the forward curve. We show how this important concept can be incorporated into the PCA in section 8.6.3.

A natural question to ask is how many factors do we need to realistically model the evolution of the forward curve? Principal component analysis can also give us insights into this and in section 8.6 we return to this issue.

The implication of the discussion of this section is that to effectively describe the evolution of the energy forward curve we need more than a single factor. We can amend the model described by equation (8.2) by adding additional sources of risk and volatility functions. For example, a 3 factor model is given by;

$$\frac{dF(t, T)}{F(t, T)} = \sigma_1(t, T)dz_1(t) + \sigma_2(t, T)dz_2(t) + \sigma_3(t, T)dz_3(t) \quad (8.3)$$

where the volatility functions $\sigma_1(\cdot)$, $\sigma_2(\cdot)$, and $\sigma_3(\cdot)$ would describe the “shifting”, “twisting”, and “bending” functions in figure 8.8, with $dz_1(\cdot)$, $dz_2(\cdot)$, and $dz_3(\cdot)$ representing independent sources of uncertainty (Brownian motions). We show in detail how to empirically estimate the volatility functions in section 8.6.

8.4 A General Multi-Factor Model of the Forward Curve

For a general multi-factor model the behaviour of the forward curve can be represented by the following equation;

$$\frac{dF(t, T)}{F(t, T)} = \sum_{i=1}^n \sigma_i(t, T) dz_i(t) \quad (8.4)$$

In this formulation there are n independent sources of uncertainty which drive the evolution of the forward curve. Each source of uncertainty has associated with it a volatility function which determines by how much, and in which direction, that random shock moves each point of the forward curve. The $\sigma_i(t, T)$ are therefore the n volatility functions associated with the independent sources of risk $dz_i(t)$. In practice we would usually set $n = 1, 2$, or 3 .

8.5 Relationship Between Forward Curve and Spot Price Models

Intuitively, a model that describes the evolution of the whole forward curve is implicitly saying something about the front end of the curve, which is simply the spot energy price, and so the forward curve models of this chapter must be related to the spot price models of chapter 6. In this section we show exactly what that relationship is. We can integrate the defining stochastic differential equation (8.4) to obtain the following solution:

$$F(t, T) = F(0, T) \exp \left[\sum_{i=1}^n \left\{ -\frac{1}{2} \int_0^t \sigma_i(u, T)^2 du + \int_0^t \sigma_i(u, T) dz_i(u) \right\} \right] \quad (8.5)$$

This equation expresses the forward curve at time t in terms of its initially observed state (time 0) and integrals of the volatility functions. The spot price is just the forward contract for immediate delivery and so the process for the spot price can be obtained by setting $T = t$, i.e.

$$S(t) = F(t, t) = F(0, t) \exp \left[\sum_{i=1}^n \left\{ -\frac{1}{2} \int_0^t \sigma_i(u, t)^2 du + \int_0^t \sigma_i(u, t) dz_i(u) \right\} \right] \quad (8.6)$$

Equation (8.6) can then be differentiated to yield the stochastic differential equation for the spot price:

$$\begin{aligned} \frac{dS(t)}{S(t)} = & \left[\frac{\partial \ln F(0, t)}{\partial t} - \sum_{i=1}^n \left\{ \int_0^t \sigma_i(u, t) \frac{\partial \sigma_i(u, t)}{\partial t} du + \int_0^t \frac{\partial \sigma_i(u, t)}{\partial t} dz_i(u) \right\} \right] dt \\ & + \sum_{i=1}^n \sigma_i(t, t) dz_i(t) \end{aligned} \quad (8.7)$$

The term in square parentheses in the drift can be interpreted as being equivalent to the sum of the deterministic riskless rate of interest $r(t)$ and a convenience yield $\delta(t)$ which in general will be stochastic. Also, since the last component of the drift term involves the integration over the Brownian motions, the spot price process will, in general, be non-Markovian – i.e the evolution of the spot price will depend upon its past evolution.

One special case of the general model is the simple single factor model described by equation (8.1). Recall that for this model we have $n = 1$ and $\sigma_1(t, T) = \sigma e^{-\alpha(T-t)}$. Clewlow and Strickland (1999a) evaluate (8.7) with this volatility function and show that the resulting spot price process is given by;

$$\frac{dS(t)}{S(t)} = \left[\frac{\partial \ln F(0, t)}{\partial t} + \alpha(\ln F(0, t) - \ln S(t)) + \frac{\sigma^2}{4}(1 - e^{-2\alpha t}) \right] dt + \sigma dz(t) \quad (8.8)$$

This implies;

$$\frac{dS(t)}{S(t)} = [\mu(t) - \alpha \ln S(t)]dt + \sigma dz(t) \quad (8.9)$$

where

$$\mu(t) = \frac{\partial \ln F(0, t)}{\partial t} + \alpha \ln F(0, t) + \frac{\sigma^2}{4}(1 - e^{-2\alpha t}).$$

This single factor forward curve model is therefore just the single factor Schwartz (1997) model of chapter 6, however with a time dependent drift term. It is this term in the drift which allows the model to now fit the observed forward prices. Note also that this particular form of the forward curve volatility function results in a “Markovian” spot price process – as the dependence in the drift on the path of the Brownian motion disappears.

The relationship between the forward curve model and the spot return model also shows that the mean reverting behaviour of the spot price is directly related to the attenuation of volatility of the forward curve. Recall also from chapter 6 that by setting $\alpha = 0$ we obtain the Black (1976) model, this is therefore a special case of the general model in equation (8.4) with $\sigma(t, T) = \sigma$ and $n = 1$.

8.6 Estimation of the Volatility Functions

Although, in the previous section, we discussed special cases which arise as restrictions of the general model, the main advantage of the forward curve modelling approach is the flexibility that the user has in choosing both the number and form of the volatility functions. Similar to other derivative markets these can be chosen in one of two general ways; historically, from time series analysis; or implied from the market prices of options.

8.6.1 Historical Estimation of the Forward Curve Volatility Functions

In this section we show how to obtain the general volatility functions from historical forward curve data. The method that we describe allows the user to determine both the form of the volatility functions as well as the number of factors to use. The general multi-factor forward curve model of this chapter can be represented, after an application of Ito's lemma, in logarithmic form, as;

$$d \ln F(t, T) = -\frac{1}{2} \sum_{i=1}^n \sigma_i(t, T)^2 dt + \sum_{i=1}^n \sigma_i(t, T) dz_i(t) \quad (8.10)$$

This can be discretized for small time changes Δt as

$$\Delta \ln F(t, t + \tau_j) = -\frac{1}{2} \sum_{i=1}^n \sigma_i(t, t + \tau_j)^2 \Delta t + \sum_{i=1}^n \sigma_i(t, t + \tau_j) \Delta z_i \quad (8.11)$$

Equation (8.11) implies that changes in the natural logarithms of the forward prices with relative maturities τ_j , $j = 1, \dots, m$ are jointly normally distributed. We can compute the sample covariance matrix of these forward prices in the standard way:

$$\hat{\sigma}_{ij} = \frac{1}{N} \sum_{k=1}^N (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j) \quad (8.12)$$

where there are N samples ($k = 1, \dots, N$) of x_{ik} and x_{jk} which are defined as;

$$x_{ik} = \ln(F(t_k, t_k + \tau_i)) - \ln(F(t_k - \Delta t, t_k - \Delta t + \tau_i)) \quad (8.13)$$

and where $\bar{x}_i$, $\bar{x}_j$ are the sample means. The time interval Δt would usually be one day. The discretised volatility functions, $\sigma_i(t, t + \tau_j)$; $i = 1, \dots, n$ and $j = 1, \dots, m$, are recovered by eigenvector decomposition of the covariance matrix. The decomposition yields the set of independent factors which drive the evolution of the variables underlying the covariance matrix. It decomposes the covariance matrix which we denote by Σ , into n eigenvectors $\underline{v}_i$ and associated eigenvalues λ_i such that

$$\Sigma = \Gamma \Lambda \Gamma^T \quad (8.14)$$

where

$$\Gamma = \begin{bmatrix} v_{11} & v_{12} & \dots & v_{1n} \\ v_{21} & v_{22} & \dots & v_{2n} \\ \dots & \dots & \dots & \dots \\ v_{n1} & v_{n2} & \dots & v_{nn} \end{bmatrix} \text{ and } \Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

and where superscript T here denotes transpose. The columns of Γ are the eigenvectors. The eigenvalues represent the variances of the independent “factors” which drive the

forward points in proportions determined by the eigenvectors. The volatility functions are then obtained as $\sigma_i(t, t + \tau_j) = v_{ji}\sqrt{\lambda_i}$.

8.6.2 Example Analysis of NYMEX Crude Oil Futures

We use the methodology of the previous section to determine the volatility functions for NYMEX crude oil futures over the period from the 1 April 1998 to 31 March 2000, using the first 24 monthly contracts for the analysis. The first step is to construct a time series of forward price returns according to equation (8.13). Table 8.1 shows the returns for the first and last four observations in the data set.

Next we compute the sample covariance matrix by applying equation (8.12). Table 8.2 shows the results for the first and last four contract months

Table 8.3 summarises the results of the eigenvector decomposition. The first 24 columns of the data, “fn 01”, ..., “fn 24”, represent the eigenvectors, with the last column, headed λ , containing the eigenvalues. Each row (excluding the last column) corresponds to the indicated forward price describing how much that particular forward is influenced by the factor associated with each column.

TABLE 8.1 Returns to NYMEX Crude Oil Futures Contracts

Date	F(1m)	F(2m)	F(3m)	F(4m)	...	F(21m)	F(22m)	F(23m)	F(24m)
02/04/98	0.0128	0.0125	0.0123	0.0115	...	0.0000	0.0000	0.0000	0.0000
03/04/98	0.0158	0.0142	0.0128	0.0114	...	0.0051	0.0046	0.0046	0.0046
06/04/98	-0.0344	-0.0312	-0.0288	-0.0266	...	-0.0160	-0.0155	-0.0149	-0.0149
07/04/98	-0.0150	-0.0140	-0.0125	-0.0111	...	0.0017	0.0017	0.0017	0.0017
...	...	...	...	...	...	...	...	...	...
28/03/00	-0.0255	-0.0244	-0.0218	-0.0190	...	-0.0158	-0.0164	-0.0169	-0.0175
29/03/00	-0.0239	-0.0184	-0.0121	-0.0067	...	0.0243	0.0249	0.0255	0.0267
30/03/00	0.0094	0.0077	0.0066	0.0063	...	0.0061	0.0061	0.0062	0.0062
31/03/00	0.0075	0.0118	0.0116	0.0113	...	0.0093	0.0089	0.0085	0.0085

TABLE 8.2 Covariance Matrix For NYMEX Crude Oil Futures

	F(1m)	F(2m)	F(3m)	F(4m)	...	F(21m)	F(22m)	F(23m)	F(24m)
F(1m)	0.1356	0.1153	0.1042	0.0959	...	0.0494	0.0486	0.0464	0.0466
F(2m)	0.1153	0.1068	0.0975	0.0901	...	0.0486	0.0478	0.0457	0.0457
F(3m)	0.1042	0.0975	0.0904	0.0834	...	0.0471	0.0464	0.0445	0.0437
F(4m)	0.0959	0.0901	0.0834	0.0783	...	0.0453	0.0446	0.0428	0.0436
...	...	...	...	...	...	...	...	...	...
F(21m)	0.0494	0.0486	0.0471	0.0453	...	0.0384	0.0381	0.0369	0.0367
F(22m)	0.0486	0.0478	0.0464	0.0446	...	0.0381	0.0381	0.0367	0.0367
F(23m)	0.0464	0.0457	0.0445	0.0428	...	0.0369	0.0367	0.0399	0.0358
F(24m)	0.0466	0.0457	0.0437	0.0436	...	0.0367	0.0367	0.0358	0.0392

TABLE 8.3 Eigenvector Decomposition for NYMEX Crude Oil Futures

	fn 01	fn 02	fn 03	fn 04		fn 21	fn 22	fn 23	fn 24	λ
F(1m)	0.2926	-0.5138	-0.7253	-0.1404	...	-0.0060	-0.0048	-0.0036	0.0033	1.2388
F(2m)	0.2764	-0.3545	0.0605	0.1155	...	0.1435	0.1755	0.0194	-0.0191	0.0888
F(3m)	0.2592	-0.2612	0.1739	0.2544	...	-0.3941	-0.4317	-0.1889	0.1656	0.0107
F(4m)	0.2452	-0.1860	0.1627	-0.0388	...	0.1060	0.0955	0.5185	-0.4368	0.0060
...	...	...	...	...	...	...	...	...	...	...
F(21m)	0.1625	0.2257	-0.1309	0.2140	...	0.0047	0.3128	-0.0050	0.0706	0.0001
F(22m)	0.1609	0.2346	-0.1511	0.2075	...	0.0317	-0.2012	0.0110	-0.0214	0.0001
F(23m)	0.1554	0.2476	-0.2672	0.6795	...	-0.0467	0.0475	-0.0099	0.0050	0.0000
F(24m)	0.1568	0.2601	-0.2419	-0.2925	...	-0.0026	-0.1135	0.0033	-0.0341	0.0000

TABLE 8.4 Volatility Functions for NYMEX Crude Oil Futures

	fn 01	fn 02	fn 03	fn 04		fn 21	fn 22	fn 23	fn 24
F(1m)	0.3257	-0.1531	-0.0751	-0.0109	...	0.0000	0.0000	0.0000	0.0000
F(2m)	0.3076	-0.1056	0.0063	0.0089	...	0.0011	0.0013	0.0001	-0.0001
F(3m)	0.2885	-0.0778	0.0180	0.0197	...	-0.0030	-0.0031	-0.0008	0.0006
F(4m)	0.2729	-0.0554	0.0169	-0.0030	...	0.0008	0.0007	0.0022	-0.0015
...	...	...	...	...	...	...	...	...	...
F(21m)	0.1808	0.0673	-0.0136	0.0166	...	0.0000	0.0022	0.0000	0.0002
F(22m)	0.1791	0.0699	-0.0157	0.0161	...	0.0002	-0.0014	0.0000	-0.0001
F(23m)	0.1729	0.0738	-0.0277	0.0526	...	-0.0004	0.0003	0.0000	0.0000
F(24m)	0.1745	0.0775	-0.0251	-0.0227	...	0.0000	-0.0008	0.0000	-0.0001

By multiplying the eigenvectors by the square root of the eigenvalue we obtain the resulting volatility functions. The columns of table 8.4 indicate the result of this calculation. The volatility functions illustrated in figure 8.8 are just the first 3 columns of this table.

The eigenvalues resulting from the eigenvector decomposition tell us the importance of each eigenvector and hence the number of factors that we should typically be using in the general model. Figure 8.9 plots the eigenvalues for the NYMEX crude oil futures data (the column headed “ λ ” in table 8.3) in descending order of size. The figure shows that the first eigenvector (correspondingly volatility function) is the most important, explaining 91.1% of the total variation in the evolution of the futures curve. Together the first 2 factors explain 97.7% of the total variation with the first 3 factors explaining 98.4%. Factors 4 and above are not significant and so a 3 factor model is sufficient to explain the evolution of the NYMEX oil data over this period.

8.6.3 Incorporating Seasonality into the Volatility Functions

As we have described earlier, gas and electricity forward curves exhibit strong seasonality. Seasonality in the volatility functions can be represented as the product of a time

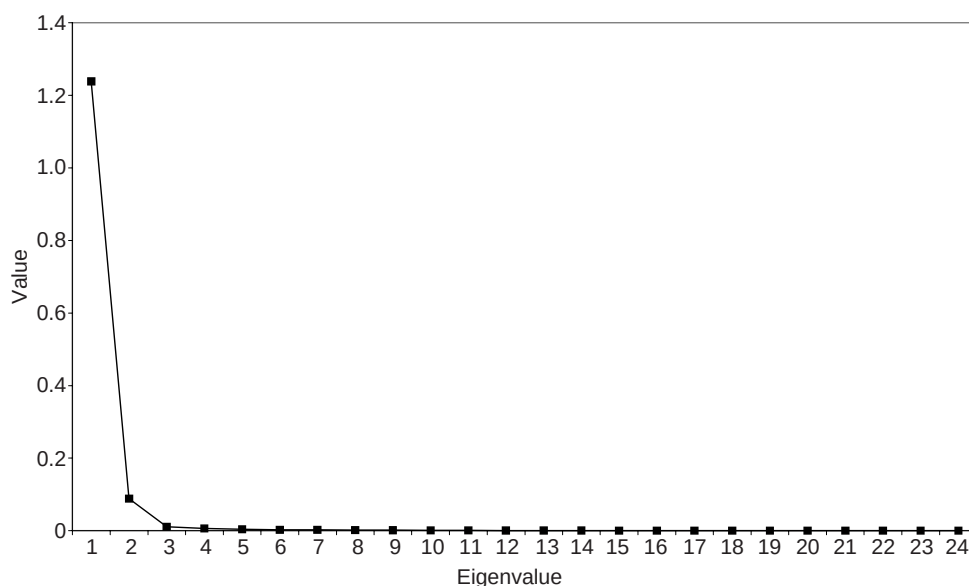


FIGURE 8.9 Eigenvalues for NYMEX Crude Oil Futures. From April 1998 to March 2000

dependent spot volatility function and maturity dependent volatility functions. The general equation (8.4) can therefore be specialised to cope with seasonality as follows:

$$\frac{dF(t, T)}{F(t, T)} = \sigma_S(t) \sum_{i=1}^n \sigma_i(T - t) dz_i(t) \quad (8.15)$$

where $\sigma_S(t)$ denotes the spot price volatility at time t and $\sigma_i(T - t)$ the n maturity dependent volatility functions. In this way, the maturity structure of the volatility functions is normalised by the spot volatility. In order to estimate this model we first estimate the spot volatility, for example, by using a rolling 30 day sample standard deviation. We then divide the daily forward price returns by the spot volatility estimate for that day – the calculations for the covariance matrix and the eigenvector decomposition are then obtained as before with this normalised set of returns.

We use the methodology of this section to determine the volatility functions for NYMEX natural gas futures contracts and NYMEX California/Oregon Border (COB) electricity futures over the period 1 April 1998 to 31 March 2000. Figure 8.10 shows the natural gas “spot” daily return standard deviation using the shortest maturity future with a rolling window of 30 days returns.

Figure 8.11 shows the result of applying the principal component analysis to the adjusted NYMEX natural gas futures data, with the first three factors shown.

Comparing figure 8.11 with figure 8.8 we see that, although the levels of the volatility functions differ between the crude oil and natural gas, qualitatively the “shifting”,

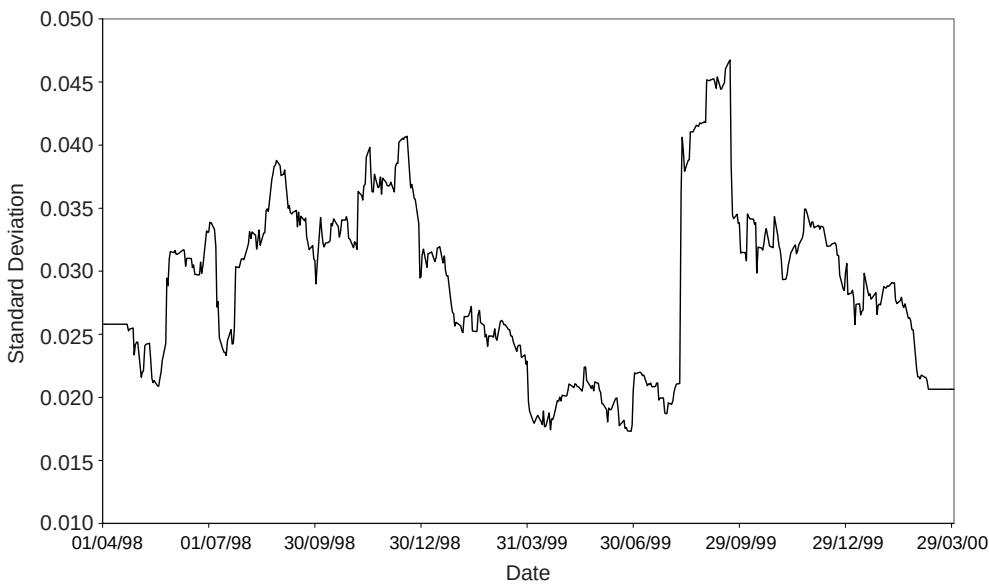


FIGURE 8.10 NYMEX Natural Gas Spot Daily Return Standard Deviation

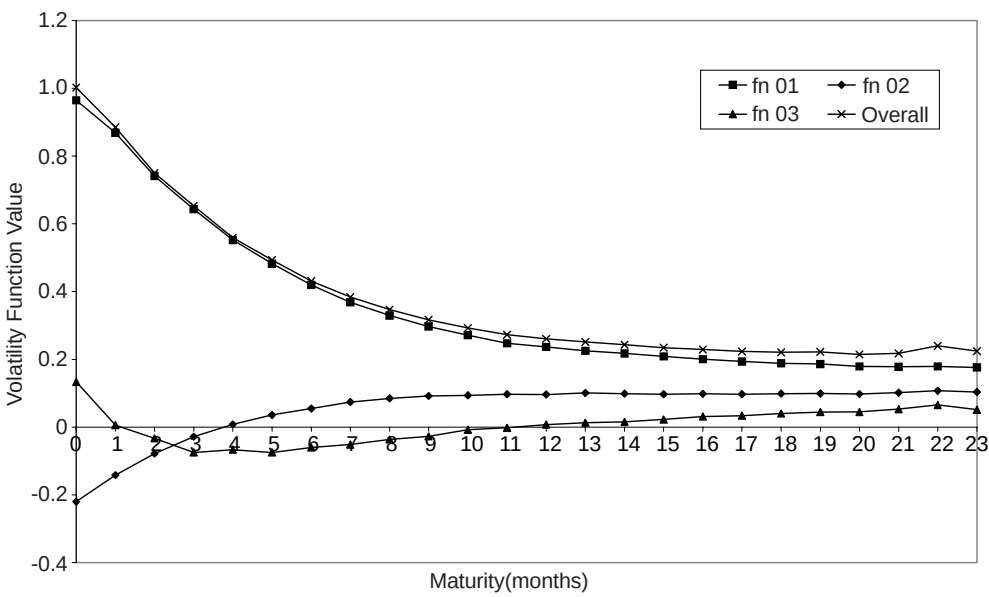


FIGURE 8.11 Seasonally Adjusted NYMEX Natural Gas Volatility Functions

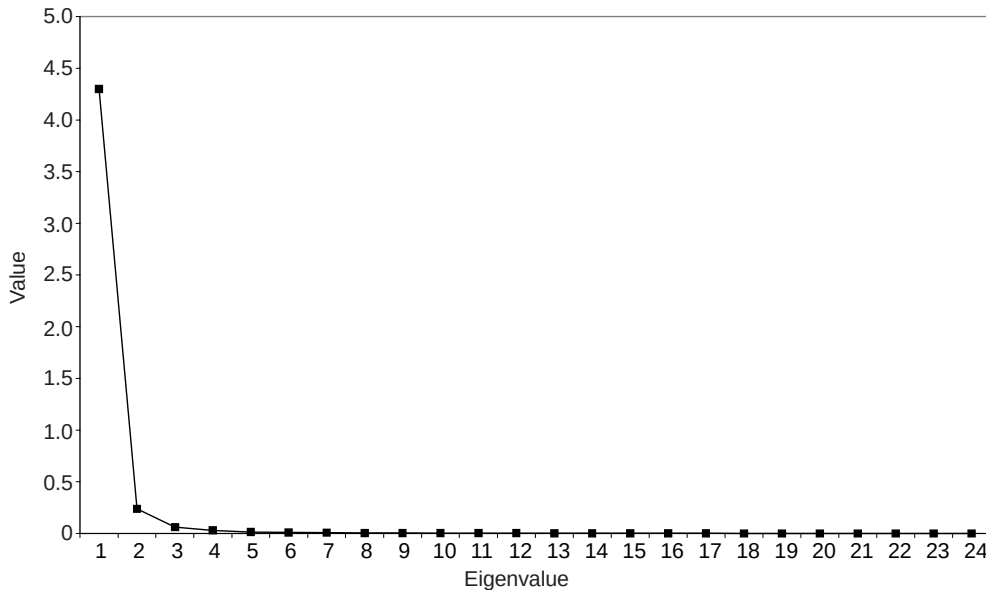


FIGURE 8.12 Eigenvalues for NYMEX Natural Gas Futures From April 1998 to March 2000

“tilting” and “bending” factors are common to both and in fact to all energy contracts that we have studied. Also notice that the zero crossing points for the tilting and bending of the curves are in different maturity segments. For example, the tilting factor crosses at 9 months maturity for oil but as early as 4 months for natural gas. Figure 8.12 shows the eigenvalues obtained via the PCA for natural gas.

Comparing figure 8.12 with figure 8.9 which shows the results of the oil analysis we see that for natural gas more factors are slightly significant. The first eigenvector now explains 91.7% of the evolution of the futures curve, the first two together 96.8%, rising to 98.1% with the addition of the third. To explain 99% we would need to use a five factor model for natural gas compared with only four factors for crude oil. Figures 8.13 and 8.14 show the analysis for NYMEX COB electricity futures.

The oscillations in the volatility functions beyond nine months maturity and the slow decline in the eigenvalues is indicative of noise in the futures prices which are fairly illiquid beyond three months maturity.

8.7 Pricing Standard European Options

In this section we turn our attention to the pricing of derivatives in the forward curve modelling framework. We begin with the pricing of standard European options on the spot energy price and then extend the analysis to pricing options on forward contracts⁴.

⁴ This section is based on section 3 of Clewlow and Strickland (1999b).

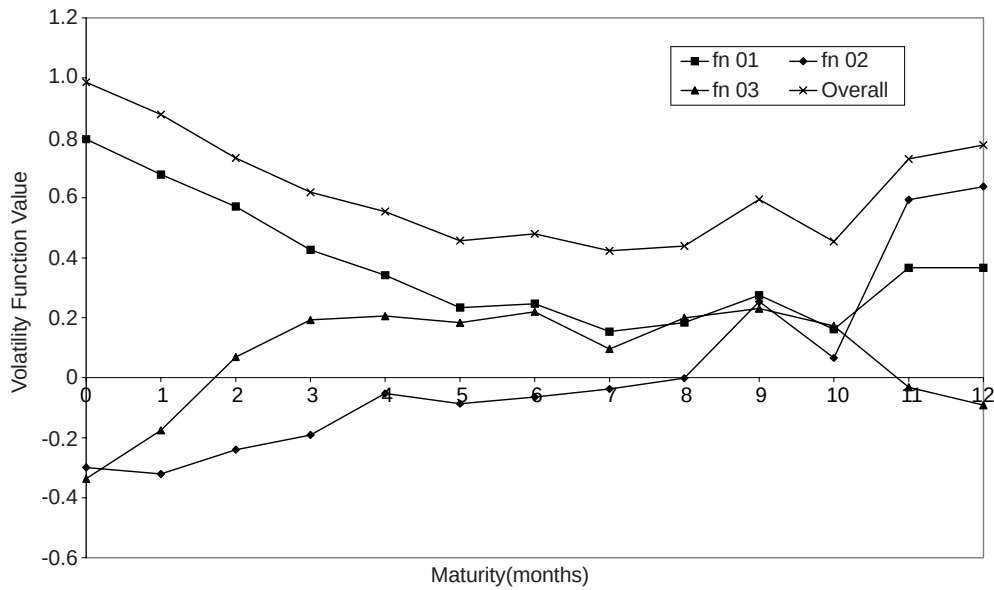


FIGURE 8.13 NYMEX COB Electricity Futures Volatility Functions From April 1998 to March 2000

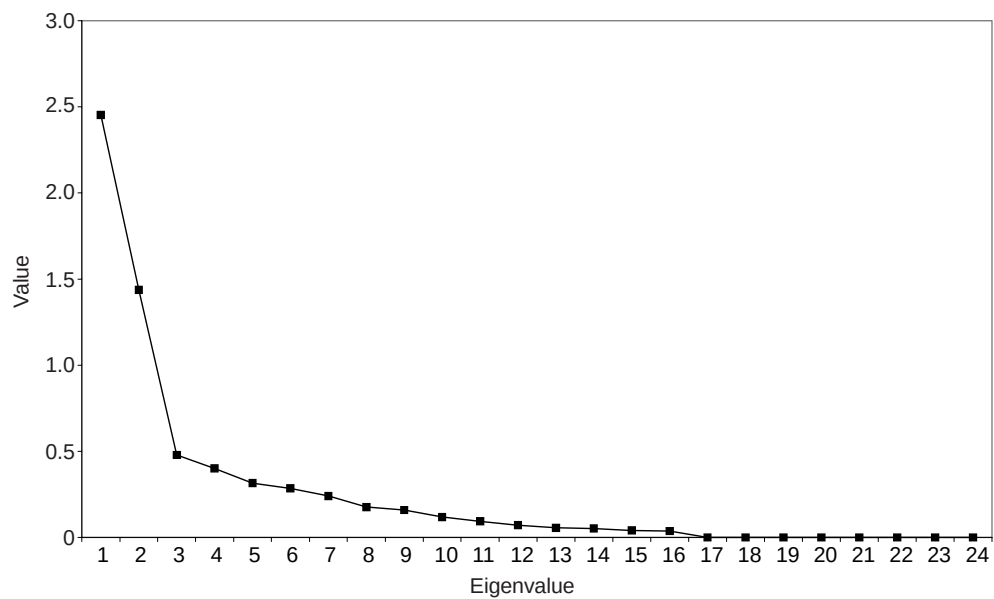


FIGURE 8.14 Eigenvalues for NYMEX COB Electricity Futures From April 1998 to March 2000

8.7.1 Options on the Spot

From standard risk-neutral pricing results the price at time t of any contingent claim on the spot energy price with payoff at time T , $C(t, S(t))$, is given by the expectation of the discounted payoff under the risk neutral measure;

$$C(t, S(t)) = E_t[P(t, T)C(T, S(T))] \quad (8.16)$$

where $P(t, T)$ is the T -maturity discount factor. For a standard European call option with strike price K and maturity date T on the spot energy price, $c(t, S(t); K, T)$, we have

$$c(t, S(t); K, T) = E_t[P(t, T) \max(0, S(T) - K)] \quad (8.17)$$

From equation (8.6) the natural logarithm of the spot price at time T , given the forward curve at time t , is normally distributed as follows:

$$\ln S(T) \sim N \left[\ln(F(t, T)) - \frac{1}{2} \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, T)^2 du \right\}, \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, T)^2 du \right\} \right] \quad (8.18)$$

The distribution for the spot energy price is therefore determined solely by the current forward price for the future date T and the assumptions regarding the volatility functions, $\sigma_i(\cdot)$. Since interest rates are assumed to be deterministic and spot prices are lognormally distributed, we can use the results of Black and Scholes (1973) to obtain the following analytical formula for a standard European call option;

$$c(t, S(t); K, T) = P(t, T)[F(t, T)N(h) - KN(h - \sqrt{w})] \quad (8.19)$$

where

$$h = \frac{\ln(F(t, T)/K) + \frac{1}{2}w}{\sqrt{w}}, \quad w = \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, T)^2 du \right\}$$

The corresponding pricing formula for a standard European put option, $p(t, S(t); K, T)$, can be easily obtained by put-call parity;

$$p(t, S(t); K, T) = P(t, T)[KN(-h + \sqrt{w}) - F(t, T)N(-h)] \quad (8.20)$$

For the single factor restriction of section 8.2, equation (8.18) can be evaluated explicitly with the distribution of the spot price satisfying;

$$\ln S(T) \sim N \left[\ln F(t, T) - \frac{\sigma^2}{4\alpha} [1 - e^{-2\alpha(T-t)}], \frac{\sigma^2}{2\alpha} [1 - e^{-2\alpha(T-t)}] \right] \quad (8.21)$$

European call and put options are then given by equations (8.19) and (8.20) respectively

with w replaced by

$$w = \frac{\sigma^2}{2\alpha} \left[1 - e^{-2\alpha(T-t)} \right] \quad (8.22)$$

A special case of equation (8.2) is the case of $\sigma(t, T) = \sigma$. This is the restriction of Amin, et al. (1995), and in this case $w = \sigma^2(T - t)$.

8.7.2 Options on Forward Contracts

We now derive the price at time t of a European call option with strike price K that matures at time T on a forward contract that matures at time s , which we denote by $c(t, F(t, s); K, T, s)$. Options are again priced using the standard methods. At date t the European call option price is the expected discounted payoff;

$$c(t, F(t, s); K, T, s) = E_t[P(t, T) \max(0, F(T, s) - K)] \quad (8.23)$$

From equation (8.5) the natural logarithms of the forward prices at time T are normally distributed;

$$\ln F(T, s) \sim N \left(\ln F(t, s) - \frac{1}{2} \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, s)^2 du \right\}, \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, s)^2 du \right\} \right) \quad (8.24)$$

Using this result it is straightforward to show that the solution to equation (8.23) is given by;

$$c(t, F(t, s); K, T, s) = P(t, T) [F(t, s)N(h) - KN(h - \sqrt{w})] \quad (8.25)$$

where

$$h = \frac{\ln(F(t, s)/K) + \frac{1}{2}w}{\sqrt{w}}, \quad w = \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, s)^2 du \right\}$$

The formula for standard European put options can be obtained by put-call parity.

Clelow and Strickland (1999a) show that for the single factor restriction of this chapter European call futures option prices are calculated via equation (8.25) with w given by;

$$\begin{aligned} w &= \int_t^T \sigma(u, s)^2 du \\ &= \int_t^T \sigma^2 e^{-2\alpha(s-u)} du \\ &= \frac{\sigma^2}{2\alpha} \left(e^{-2\alpha(s-T)} - e^{-2\alpha(s-t)} \right) \end{aligned} \quad (8.26)$$

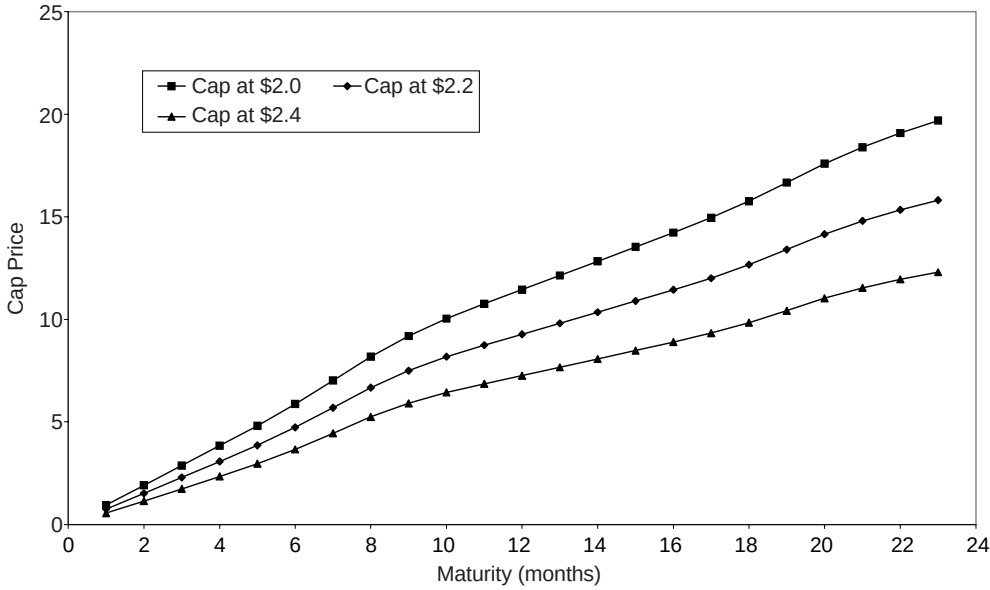


FIGURE 8.15 Natural Gas Cap Prices with Monthly Maturities

Armed with these analytic results it is possible to price other standard energy derivatives. For example, in Chapter 5 we showed that an energy cap can be viewed as a portfolio of call options on futures contracts. Figure 8.15 shows the prices of caps with monthly maturities from 1 month until 23 months for natural gas under the general model (see equation (5.3)). Each cap price is the sum of the monthly caplets up to the cap maturity. The individual caplets have been priced using the observed NYMEX forward curve for natural gas for 31 March 2000 (illustrated in figure 8.1) and the three volatility functions derived from our historical analysis (illustrated in figure 8.11).

8.8 Pricing General European Contingent Claims

In this section we show how to price general European contingent claims for the multi-factor forward curve model described by equation (8.4). For example, consider a European option, with maturity T , on a portfolio of forward prices with maturities on a set of dates $s_k, k = 1, \dots, m$ ($T \leq s_1 \leq \dots \leq s_m$) with face values c_k . The payoff at the maturity date T is defined to be $C(T, \{c_k, F(T, s_k)\})$ with the price at time t given by the expectation;

$$C(t, \{c_k, F(t, s_k)\}) = E_t[P(t, T)C(T, \{c_k, F(T, s_k)\})] \quad (8.27)$$

Recall that $E_t[\cdot]$ is the expectation operator which can be evaluated via Monte Carlo simulation simply as the average of the simulated payoff values. Derivative valuation in

this section therefore involves simulating the observed forward price curve $F(t, s_k)$ until time T to obtain $F(T, s_k)$ for all k . The expectation of equation (8.27) is then taken over the m -dimensional normal distribution of the correlated natural logarithms of the forward prices, $\ln(F(T, s_k))$. In order to perform Monte-Carlo simulation we compute the $m \times m$ covariance matrix of the forward prices at the maturity of the option, Σ , in the standard way;

$$\Sigma_{kl} = \text{Cov}[\ln(F(T, s_k)), \ln(F(T, s_l))] = \sum_{i=1}^n \left\{ \int_t^T \sigma_i(u, s_k) \sigma_i(u, s_l) du \right\} \quad (8.28)$$

To efficiently sample under this covariance matrix we compute the orthogonal representation of the covariance matrix (or PCA) which returns the m eigenvectors $\underline{w}_i$ and associated m eigenvalues λ_i such that

$$\Sigma = \Gamma \Lambda \Gamma^T \quad (8.29)$$

where

$$\Gamma = \begin{bmatrix} w_{11} & w_{12} & \dots & w_{1m} \\ w_{21} & w_{22} & \dots & w_{2m} \\ \dots & \dots & \dots & \dots \\ w_{m1} & w_{m2} & \dots & w_{mm} \end{bmatrix} \text{ and } \Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_m \end{bmatrix}$$

Let M be the number of simulations and $\varepsilon_i, i = 1, \dots, m$ be independent standard normal random numbers. Clewlow and Strickland (1999b) show that the expectation in equation (8.27) can be represented by;

$$C(t, \{c_k, F(t, s_k)\}) = P(t, T) \frac{1}{M} \sum_{j=1}^M [C(T, \{c_k, F(t, s_k) y_j(t, T, s_k); k = 1, \dots, m\})] \quad (8.30)$$

where

$$Y_j(t, T, s_k) = \exp \left[-\frac{1}{2} \sum_{i=1}^m \{w_{ki}^2 \lambda_i\} + \sum_{i=1}^m \{w_{ki} \sqrt{\lambda_i} \varepsilon_i\} \right]$$

Notice that because of the Gaussian nature of our framework we effectively jump straight to the end of the life of the option under each simulation, rather than the more usual practice of simulating at a large number of small discrete time steps until maturity. This formulation also gives a natural way to trade off accuracy with speed by truncating the integration space dependent on the sizes of the eigenvalues. The size of eigenvalues indicate the relative importance of the corresponding eigenvector factors in reproducing

the covariance matrix Σ . Therefore by truncating the eigenvector representation we reduce the dimensionality of the Monte Carlo integration and thus reduce the computation time required but at a cost of reducing the accuracy.

Example: Pricing a European Energy Swaption

Recall from section 5.5 that the time t value of an energy swaption, with maturity date T , to swap a series of floating spot price payments on dates s_k , $k = 1, \dots, m$, for a fixed strike price amount K can be written as:

$$Swpn(t; K, T, \{s_k\}, m) = P(t, T) E_t \left[\max \left(0, \left\{ \frac{1}{m} \sum_{k=1}^m F(T, s_k) \right\} - K \right) \right] \quad (8.31)$$

From equation (8.30), we have;

$$\begin{aligned} Swpn(t; K, T, \{s_k\}, m) &= P(t, T) \frac{1}{M} \sum_{j=1}^M \left[\max \left(0, \frac{1}{m} \sum_{k=1}^m F(T, s_k) - K \right) \right] \\ &= P(t, T) \frac{1}{M} \sum_{j=1}^M \left[\max \left(0, \frac{1}{m} \sum_{k=1}^m F(t, s_k) Y_j(t, T, s_k) - K \right) \right] \end{aligned} \quad (8.32)$$

where the exponential martingale $Y_j(t, T, s_k)$ is defined in equation (8.30) and the ε_i , $i = 1, \dots, m$ are new samples for each j .

Table 8.5 shows the prices of swaptions on natural gas, using the same gas data (forward curve and volatilities) as for the previous cap example. The maturity of the options range from 1 month to 1 year. Each of these options exercises into swaps with maturities of 1, 3, or 6 months ('Swap Tenor') – each swap has monthly resets. All swaptions have an exercise price of $K = \$2.20$. Notice that, because of the seasonality observed in natural gas prices, option prices do not necessarily increase with either option maturity or swap tenor.

TABLE 8.5 Example Natural Gas Swaption Prices

		Swap Tenor		
		1m	3m	6m
Option Maturity	1m	0.7528	0.7633	0.7872
	3m	0.7784	0.7753	0.8549
	6m	0.8709	0.9280	0.7954
	1yr	0.5316	0.5182	0.5117

8.9 Pricing Exotic Options

In this section we extend the analysis of the previous sections to pricing a range of exotic or path dependent options under the multi-factor forward curved model of this chapter. We begin by pricing the two types of spread options described in section 5.7. We also consider the pricing of Asian options and American style options in this section⁵.

8.9.1 Calendar Spread Options

Recall from section 5.7.1 that calendar spread options are written on the difference between futures contracts with different maturity dates on the same underlying energy with payoff condition given by equation (5.9). Comparing equation (5.9) with equation (8.27) we see that the pricing of a calendar spread option fits in with the general pricing methodology of section 8.8. Therefore we can use equation (8.30) to evaluate the solution to the expectation;

$$\begin{aligned}
 & \text{Calendar_Spread}(t; K, T, s_1, s_2) \\
 &= P(t, T) \frac{1}{M} \sum_{j=1}^M [\max(0, F(T, s_1) - F(T, s_2) - K)] \\
 &= P(t, T) \frac{1}{M} \sum_{j=1}^M [\max(0, F(t, s_1) Y_j(t, T, s_1) - F(t, s_2) Y_j(t, T, s_2) - K)]
 \end{aligned} \tag{8.33}$$

8.9.2 Crack Spread Options

Recall from section 5.7.2 that if the futures contracts underlying the option are written on two separate energies (say natural gas and electricity), then the option is referred to as a crack spread option.

Because option contracts of this type depend simultaneously on forwards related to separate energies then, in order to value options of this type, we require the joint distribution of the two forward price curves. We can extend the multi-factor forward model of equation (8.4) to simultaneously model a number of different energy forward curves⁶. The following stochastic differential equation describes the joint evolution of energy forward curves (indexed by e) for an n -factor model;

$$\frac{dF_e(t, T)}{F_e(t, T)} = \sum_{i=1}^n \sigma_{ei}(t, T) dz_i(t) \tag{8.34}$$

⁵ See chapter 5 for definitions and examples of these and other path dependent derivatives.

⁶ This methodology is used in chapter 10 for VaR calculations for energy portfolios that contain multiple energies.

where e = oil, gas, electricity, etc. Increments in the different energy forward curves are expressed in terms of a consistent set of volatility functions, $\sigma_{ei}(t, T)$, which describe how all the forward curves evolve together. For example, the first volatility function describes how the primary factor affects all the energy forward curves. Under this model all forward price returns are jointly normally distributed. Option prices are again computed via Monte Carlo simulation. In order to simulate the joint behaviour of the energy prices on which the portfolio depends we must compute the volatility functions $\sigma_{ei}(t, T)$ from the covariance matrix of all the relevant forward prices. An example of this joint estimation is shown in chapter 10 for calculating value-at-risk for multiple energy portfolios.

If $F_a(t, s_1)$ represents the price of an s_1 -maturity futures contract on energy a and $F_b(t, s_2)$ represents the price of an s_2 -maturity futures contract on energy b then the price of the option, as represented by equation (5.12), can be evaluated as follows;

$$\begin{aligned} & Crack_Spread(t; K, T, F_a(t, s_1), F_b(t, s_2), s_1, s_2) \\ &= P(t, T) \frac{1}{M} \sum_{j=1}^M [\max(0, F_a(T, s_1) - F_b(T, s_2) - K)] \\ &= P(t, T) \frac{1}{M} \sum_{j=1}^M [\max(0, F_a(t, s_1) Y_j(t, T, s_1) - F_b(t, s_2) Y_j(t, T, s_2) - K)] \end{aligned} \quad (8.35)$$

Note that the calculation of the exponential martingales now depends on the elements of the decomposed *joint* covariance matrix derived from the evolution of the two energy forward curves.

8.9.3 Asian Options

As an example of an option with a terminal payoff dependent on the path taken by the spot instrument or a forward contract, valued in the forward curve framework, we show how to price an Asian option. The methodology we describe is equally suited to barriers, lookbacks, and other exotic options. We consider the following fixed strike European Asian call option. We assume the maturity of the option to be s_m with the payoff determined by the average of the s_m -maturity futures contract observed at times $s_1, \dots, s_m$ with $s_1 < \dots < s_m$. The last observation is therefore also the spot price at time s_m . This option is an example of the average price call option in table 5.1 and so the value can be written as the following discounted expected payoff;

$$Asian(t; K, s_m, \{s_k\}) = P(t, s_m) E_t \left[\max \left(0, \left\{ \frac{1}{m} \sum_{k=1}^m F(s_k, s_m) \right\} - K \right) \right] \quad (8.36)$$

Note that, if the average were taken over the futures contracts with a constant time to maturity Δs (for example 1 month) on each observation date the payoff would be

written as;

$$Asian(t; K, s_m, \{s_k\}) = P(t, s_m) E_t \left[\max \left(0, \left\{ \frac{1}{m} \sum_{k=1}^m F(s_k, s_k + \Delta s) \right\} - K \right) \right] \quad (8.37)$$

In order to evaluate the expectation in equation (8.36) (or 8.37) we need a more general representation than (8.27) because the maturity payoff is determined by observables at times other than maturity. We now consider any contingent claim to be a series of possibly contingent cashflows, $C_k(s_k, \Theta, \{c_l F(s_k, s_l)\})$, occurring on dates $s_k, k = 1, \dots, m$ and depending on forward prices with maturity dates also on the set of dates s_k with face values c_k and also on the parameter vector Θ . The price of a contingent claim of this form is given by

$$C(t, \Theta, \{c_k\}, \{s_k\}) = E_t \left[\sum_{k=1}^m P(t, s_k) C_k(s_k, \Theta, \{c_l F(s_k, s_l)\}) \right] \quad (8.38)$$

The expectation is taken over the $m \times m$ -dimensional normal distribution of the correlated natural logarithms of the forward prices $\ln(F(s_k, s_l))$. In order to perform Monte-Carlo simulation to evaluate (8.38) we must therefore compute the $m' \times m'$ covariance matrix Σ (where $m' = m^2$);

$$\Sigma_{jk} = Cov[\ln(F(t_j, s_j)), \ln(F(t_k, s_k))] = \sum_{i=1}^n \left\{ \int_t^{\min(t_j, t_k)} \sigma_i(u, s_j) \sigma_i(u, s_k) du \right\} \quad (8.39)$$

In order to efficiently sample under this covariance matrix we again compute the orthogonal representation of the covariance matrix which now yields the m' eigenvectors $\underline{w}_i$ and associated m' eigenvalues λ_i . Using this result, equation (8.38) can be written as;

$$C(t, \Theta, \{c_k\}, \{s_k\}) = \frac{1}{M} \sum_{j=1}^M \left[\sum_{k=1}^m P(t, s_k) C_k(s_k, \Theta, \{c_l F(t, s_l) Y_j(t, s_k, s_l)\}) \right] \quad (8.40)$$

where

$$Y_j(t, s_k, s_l) = \exp \left[-\frac{1}{2} \sum_{i=1}^{m'} \{w_{il'}^2 \lambda_i\} + \sum_{i=1}^{m'} \{w_{il'} \sqrt{\lambda_i} \varepsilon_i\} \right], l' = kl$$

Applying (8.40) to our example of the Asian option of equation (8.36) we therefore have;

$$Asian(t; K, s_m, \{s_k\}) = P(t, s_m) \frac{1}{M} \sum_{j=1}^M \max \left(0, \left\{ \frac{1}{m} \sum_{k=1}^m F(t, s_m) Y_j(t, s_k, s_m) \right\} - K \right) \quad (8.41)$$

Other path dependent options can similarly be handled in this way. For example, for European barrier options with the barrier fixed against the level of the s_m -maturity futures contract observed at dates $s_1, \dots, s_m$, we can use the above methodology to

calculate the forward observations $F(s_k, s_m)$ for $k = 1, \dots, m$. The values are then compared to the barrier level to determine if the barrier has been crossed⁷. Similarly for lookback options, the forwards on the reset dates can be calculated using this method and the maximum or minimum determined from these observations dependent on the option characteristics⁸.

8.9.4 American Options in Forward Curve Models

Although American options are not usually considered exotic (unless we are looking at the early exercise of some of the path dependent options already considered in this section) their values are typically difficult to evaluate in the simulation setting of this chapter. The problem arises because simulation methods generate trajectories of state variables forward in time, whereas a backward dynamic programming approach is required to efficiently determine optimal decisions for pricing American options. We outline in this section how, by supplementing the Monte Carlo integration of previous sections with Markovian spot price trees in order to generate an early exercise strategy, we can efficiently obtain approximations for prices of American style derivatives. The price of an American style contingent claim reflects the extra premium derived from the ability to exercise early. Equation (8.38) can be further generalised to reflect the optimal choice of early exercise strategy by allowing the parameter vector Θ to now include the early exercise strategy.

The single factor model of section 7.3 can be considered a one factor Markovian restriction of the general model of this chapter. Recall from chapter 7 (equation (7.27)) that the logarithm of the spot price follows the process;

$$dx(t) = [\theta(t) - \alpha x(t)]dt + \sigma dz(t) \quad (8.42)$$

where the time dependent function is determined by the initial forward price curve. A trinomial tree was constructed in section 7.3, consistent with both the current forward price and forward volatility curves. Similarly, we can construct a trinomial tree consistent with the initial forward curve and the overall volatility function in the multi-factor model. The trinomial tree constructed in this way allows us to determine an approximate early exercise strategy for American contingent claims. We use the approximate early exercise strategy obtained in this way as an input into our multi-factor non-Markovian Monte Carlo simulation. If we represent this early exercise strategy by the parameter vector $\tilde{\Theta}$ we can obtain a lower bound approximation to the price of the American contingent claim under the full model using

$$\tilde{C}(t, \tilde{\Theta}, \{c_k\}, \{s_k\}) = E_t \left[\sum_{k=1}^m P(t, s_k) C_k(s_k, \tilde{\Theta}, \{c_l F(s_k, s_l)\}) \right] \quad (8.43)$$

⁷ See chapter 5 for a description of barrier options and a definition of typical option payoffs.

⁸ See chapter 5 for a description of lookback options and a definition of typical option payoffs.

Furthermore we can obtain an arbitrarily good approximation by maximising the value of the contingent claim with respect to the early exercise strategy.

8.10 Implied Estimation of the Volatility Functions

Many practitioners argue that the most important issue in the use of any model for pricing and hedging derivatives is that of calibrating the model to market data. Calibration is the process of choosing model parameters so that the prices returned by the model coincide with the observed market prices. Calibrating an energy model is analogous to choosing, or implying, a volatility parameter for the Black-Scholes-Merton model, when valuing say stock or index options, that equates the model price with the market price. However, in the Black-Scholes-Merton model a single variable carries all the volatility information, but in forward curve models it is the set of volatility functions. Exact implied calibration to all market prices simultaneously is often not possible, depending on the volatility functions chosen, and so the procedure usually involves choosing the model parameters so that prices returned by the model are as close as possible to the market prices of options. This closeness of fit is often measured in a ‘least-squares’ sense. Define the set of market prices to be $C_{market,i}$; $i = 1, \dots, N$ and the corresponding model prices be $C_{model,i}(\Theta)$ where Θ is the set of parameters of the option pricing model. To calibrate the model prices to the market prices we solve the following problem:

$$\text{minimise } \left\{ \sum_{i=1}^N \left(\frac{C_{market,i} - C_{model,i}(\Theta)}{C_{market,i}} \right)^2 ; \Theta \right\} \quad (8.44)$$

For clarity we will illustrate the implied estimation process with the single factor model of equation (8.1), but the extension to the general multi-factor model is straightforward. The data we use for our calibration exercise is Brent crude oil futures options prices traded on the International Petroleum Exchange (IPE) in London for 15 July 1999. Although these options have American style exercise, we assume for this example they have European style exercise, so that we can use equation (8.25) and the corresponding put option pricing formula.

The first section of table 8.6, “Market Prices”, shows the prices for Brent crude oil call and put futures options for maturities between September 1999 and February 2000, and for strike prices ranging from \$13.00 to \$21.00 on the 15 July 1999. The maturity dates for both the options and the futures contracts were obtained from the IPE calendar, with the future expiring approximately 1 week after the option maturity. Missing prices imply that the option did not trade on the 15 July 1999.

The second section of table 8.6, “Model Prices”, shows the model prices obtained via equation (8.25) and the corresponding put price. Recall that model prices are determined once the volatility curve is specified, which for this model requires specifying α and σ (the vector Θ in equation 8.26). We employed a standard numerical search routine to find α and σ by minimising the squared percentage errors across all of the options in table 8.5.

TABLE 8.6 Calibration of the Schwartz One Factor Model to IPE Brent Crude Oil Futures Options Prices

Strike Price	Market Prices											
	Calls						Puts					
	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00
	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00
13.00							0.00	0.03				
13.50	5.58						0.00					
14.00							0.00	0.00	0.04	0.06	0.14	
14.50							0.00	0.01	0.06	0.09	0.20	
15.00			3.97				0.01	0.02	0.10	0.13	0.28	
15.50			3.52				0.02	0.03				
16.00				3.00			0.04	0.06		0.27	0.51	
16.50	2.67						0.07	0.11				
17.00	2.20	2.17	2.28	2.27			0.12	0.19				
17.50	1.77			1.95	2.03	1.76	0.19	0.31		0.72	0.98	0.88
18.00	1.38	1.44	1.59	1.67	1.74	1.49	0.30	0.46	0.72	0.94	1.19	1.11
18.50	1.02	1.14	1.29	1.42	1.47	1.25	0.44	0.66	0.92	1.19	1.42	1.37
19.00	0.72	0.90	1.06	1.20	1.25	1.04	0.64	0.92	1.19	1.47	1.70	1.66
19.50	0.50	0.70	0.86	1.01	1.05	0.86	0.92	1.22	1.49	1.78	2.00	1.98
20.00	0.37	0.54	0.70	0.85	0.88		1.29	1.56	1.83			
20.50		0.40										
21.00		0.29										
Strike Price	Model Prices											
	Calls						Puts					
	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00
	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00	Sep-99	Oct-99	Nov-99	Dec-99	Jan-00	Feb-00
13.00	6.01	5.84	5.67	5.48	5.26	5.06	0.00	0.00	0.01	0.02	0.04	0.06
13.50	5.52	5.36	5.19	5.02	4.81	4.63	0.00	0.00	0.02	0.03	0.06	0.09
14.00	5.02	4.87	4.73	4.56	4.38	4.21	0.00	0.01	0.03	0.06	0.10	0.14
14.50	4.53	4.39	4.27	4.12	3.96	3.80	0.00	0.02	0.05	0.09	0.15	0.20
15.00	4.04	3.92	3.82	3.70	3.55	3.42	0.00	0.03	0.09	0.14	0.22	0.28
15.50	3.55	3.46	3.39	3.29	3.17	3.06	0.01	0.06	0.14	0.21	0.30	0.38
16.00	3.06	3.01	2.97	2.90	2.81	2.71	0.02	0.10	0.21	0.30	0.41	0.50
16.50	2.59	2.59	2.59	2.54	2.47	2.39	0.04	0.17	0.30	0.42	0.55	0.65
17.00	2.14	2.19	2.22	2.21	2.16	2.10	0.09	0.26	0.42	0.56	0.70	0.82
17.50	1.72	1.83	1.89	1.90	1.88	1.83	0.16	0.38	0.57	0.73	0.89	1.01
18.00	1.34	1.50	1.59	1.62	1.62	1.59	0.28	0.54	0.76	0.92	1.10	1.24
18.50	1.01	1.21	1.33	1.37	1.38	1.37	0.44	0.74	0.97	1.15	1.34	1.48
19.00	0.74	0.96	1.10	1.15	1.18	1.17	0.66	0.98	1.22	1.41	1.60	1.75
19.50	0.52	0.75	0.89	0.96	1.00	1.00	0.93	1.26	1.50	1.69	1.89	2.04
20.00	0.35	0.58	0.72	0.79	0.84	0.85	1.26	1.57	1.81	2.00	2.20	2.35
20.50	0.23	0.44	0.58	0.65	0.70	0.72	1.63	1.92	2.15	2.34	2.53	2.69
21.00	0.14	0.32	0.46	0.53	0.58	0.60	2.04	2.30	2.51	2.69	2.88	3.03

In this way the calibration is a ‘best fit’ to the full set of market data. The values found in this way yield $\alpha = 0.741$ and $\sigma = 0.355$. In practice, we would potentially fit the model to the prices of other energy derivatives at the same time. For example, simultaneously fitting to oil swaptions data would enable us to better fit the covariance structure of the forward contracts.

8.10.1 Handling Volatility Skews and Smiles

The importance of the concept of the implied volatility smile was introduced in chapter 1 and its incorporation into spot price models has been discussed in chapters 2, 3, 6 and 7. Incorporating volatility smile information into the multi-factor forward curve models of this chapter can also be achieved with varying levels of complexity.

Probably the simplest approach is via a simple extension to the calibration procedure in the previous section. Once the volatility function(s) have been calibrated we simply compute the required volatility scaling factor for each strike price and option maturity which makes the model price exactly equal to the market price. Interpolation between these scaling factors can then be used to price options with strikes and maturities not observable in the market. However, this approach does not help in the pricing of exotic options consistent with the market prices of the standard options.

A simple way to price exotic options is to compute the value of the exotic option in a trinomial tree fitted to the multi-factor model as described in section 8.9.4 and in a implied trinomial tree also fitted to the market prices of standard options. The difference between these two prices can be taken to be the pricing differential due to the volatility smile. This can then be added to the price obtained for the exotic under the full multi-factor model using the techniques in section 8.9.

8.11 Summary

In this chapter we have developed a general framework for the pricing and risk management of energy derivatives based on the observed forward curve for energy prices. The framework is designed to be consistent not only with the market observable forward price curve but also the volatilities and correlations of forward prices. We have shown that there is a direct relationship between the forward curve models and the spot price models of chapter 6. Estimation of the volatility functions of the forward curve model from both historical and cross-sectional market data was discussed including the incorporation of seasonality and volatility smiles. Although based on very general volatility functions, we develop closed-form solutions for standard European options on the spot energy price and forward prices. For options not amenable to analytical solutions we also develop a general Monte Carlo framework for pricing a wide range of European style energy options. The approach can also handle the pricing of exotic or path dependent options, and we illustrated this with examples related to calendar and crack spreads, and Asian options. Finally, we indicated how American options can be priced using a combination of trinomial trees and Monte Carlo simulation.